

STEPGEN · DESIGN DECISION RECORD

# StepGen TRL 2 review decisions

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DOCUMENT CONTROL

|             |                                                            |         |                |
|-------------|------------------------------------------------------------|---------|----------------|
| Document ID | SGN-DDR-001                                                | Version | 0.1            |
| Status      | Draft                                                      | Date    | 2026-09-25     |
| Author      | Amish Chadha                                               | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept                                     |         |                |
| Repository  | github.com/BoujeeEnjinia1701/stepgen                       |         |                |
| Source      | docs/decisions/0001-trl2-review-decisions.md @ uncommitted |         |                |

REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                    |
|---------|------------|--------------|---------------------------------------------------------------------------------------------------------------------------|
| 0.1     | 2026-09-25 | Amish Chadha | Record Amish's 2026-09-25 decisions on the TRL 2 review, the move to SwapCell interface v0.3 and the items that stay open |

## 0001: TRL 2 review decisions

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- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 12); items 13 to 15 remain proposed, awaiting Amish

### ■ Context

The TRL 2 review note ( docs/REVIEW.md , session "concept change to a walking-treadmill vehicle") listed nine items as "Proposed, awaiting Amish", most with a recommendation. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." In the same instruction he approved three additions to the SwapCell interface (a wake method for hosts without CAN, a charge-while-discharging mode and a latch vibration rating for vehicles), a pricing rule for shared SwapCell packs, and a rule that community designs pick co-design partners per area later. He also confirmed the move of StepGen from the CleanTech area to Mobility and Logistics.

The concept change itself (a walking-treadmill vehicle instead of a stationary stepper generator) was decided by Amish on 2026-09-24 and is recorded in SGN-PRB-001, SGN-PRC-001 and the review note. It is not repeated here.

This record lists what the 2026-09-25 instruction decides, and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

### ■ Options considered

The options for each item are in docs/REVIEW.md (TRL 2 session) and SGN-PRC-001 v0.3. They are not repeated here.

### ■ Decision

*Table 1. Decided items. Each is "Decided by Amish, 2026-09-25: go with recommendation" unless the row says otherwise.*

| #  | Item                  | Decision                                                                                                                                                                                                                                                                                                                                                         | Where it now lives                          |
|----|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 1  | Form factor           | Decided by Amish, 2026-09-25: go with recommendation. Two-wheel, long-wheelbase, bike type (about 2.35 m) so that a natural stride fits on the belt; the three-wheel tadpole version stays open until co-design shows whether non-cyclists can balance                                                                                                           | SGN-PRC-001 v0.4, SGN-DWG-001               |
| 2  | Wheels                | Decided by Amish, 2026-09-25: go with recommendation. 20 in (ETRTO 406) front and rear                                                                                                                                                                                                                                                                           | SGN-PRC-001 v0.4, bom/bom.csv items 6 and 7 |
| 3  | Motor and speed class | Decided by Amish, 2026-09-25: go with recommendation. 250 W rated rear geared hub, assist cut at 25 km/h, 15 km/h beginner mode (EU pedelec limits)                                                                                                                                                                                                              | SGN-REQ-001 R3 and R4                       |
| 4  | Target range          | Decided by Amish, 2026-09-25: go with recommendation. 30 km at 20 km/h on the flat on one SwapCell pack                                                                                                                                                                                                                                                          | SGN-REQ-001 R5                              |
| 5  | Regeneration          | Decided by Amish, 2026-09-25: go with recommendation. No roller generator in the first concept; keep a mounting point on the front roller. SwapCell v0.3 mode 4 now removes the interface gap, but the decision stands                                                                                                                                           | SGN-PRC-001 v0.4                            |
| 6  | Area                  | Decided by Amish, 2026-09-25: go with recommendation, and confirmed for this session. area moves from CleanTech to Mobility and Logistics, matching SunSpoke and WaterWalker                                                                                                                                                                                     | project.yaml                                |
| 7  | Budget                | Decided by Amish, 2026-09-25: go with recommendation. budget_usd raised from \$400 to \$650, SwapCell pack excluded                                                                                                                                                                                                                                              | project.yaml, SGN-REQ-001 R12               |
| 8  | Legal route           | Decided by Amish, 2026-09-25: go with recommendation. Pick a first target country and confirm the vehicle's legal category there before any road use. Which country is not decided (item 14)                                                                                                                                                                     | SGN-PRB-001 v0.4, SGN-REQ-001               |
| 9  | SwapCell interface    | Decided by Amish, 2026-09-25 (cross-cutting): StepGen builds to SwapCell interface v0.3. It uses item W (10 kΩ coding resistor in the INTERLOCK loop, which wakes the pack), fits a receiver to item V (latch class V1: 1.72 kN pack latch proof, 330 N or more receiver preload) and does not use item C (charge-discharge mode) because no generator is fitted | SGN-REQ-001 R11 and R13, SGN-PRC-001 v0.4   |
| 10 | Shared pack pricing   | Decided by Amish, 2026-09-25 (cross-cutting): the SwapCell pack is priced once, in the SwapCell BOM, and excluded from the StepGen budget                                                                                                                                                                                                                        | bom/bom.csv item 11, bom/bom-notes.md       |
| 11 | Co-design partner     | Decided by Amish, 2026-09-25 (cross-cutting): community designs pick co-design partners per area later. The partner stays open                                                                                                                                                                                                                                   | SGN-PRB-001 v0.4                            |
| 12 | Decision record       | Decided by Amish, 2026-09-25: go with recommendation (TRL 2 suggestion). This record is SGN-DDR-001                                                                                                                                                                                                                                                              | This file                                   |

### Items that remain open

These items had no recommendation to accept, or arose at TRL 3, and stay **Proposed, awaiting Amish**:

- **13. First user group.** Non-cycling adults, older adults or commuters. No preference was stated, and the partner is left open by item 11.
- **14. First target country** for the legal classification (item 8 decides the route, not the country).

- **15. New TRL 3 items** from SGN-CAL-001, each with options and a recommendation in `docs/REVIEW.md` (session 2026-09-25: TRL 3): the cost overrun against the new \$650 budget, belt length for tall riders, deck rail size for weld fatigue, and a key switch in the INTERLOCK loop.

## ■ Consequences

- SGN-PRB-001, SGN-PRC-001 and SGN-REQ-001 move to version 0.4 (they were already at 0.3 from the 2026-09-25 concept change) and no longer mark items 1 to 12 as proposed.
- R12 now reads \$650. The priced BOM is about \$713, so R12 is **not met** (SGN-CAL-001); options are in the review note.
- R11 now cites SwapCell interface v0.3, and a new requirement R13 covers pack retention to latch class V1.
- TRL 3 is the hard stop. TRL 4 (a test article, lab tests and a build log) is on hold by Amish's instruction.